

Question:

At birth, the least developed part of the brain is the

Answer Choices:

A. visual system

B. thalamus

C. frontal lobe

D. brain stem

E. parietal lobe

F. cerebellum

G. cortex

H. basal ganglia

I. hypothalamus

J. limbic system

Reasoning + Answer:

<tool_call> reasoning process here

At birth , the brain undergoes rapid development , with different regions developing at different rates . While the brain stem , hypothalamus , and other subcortical structures are present and functional from birth , the most complex and highest-level areas of the brain , particularly the cerebral cortex , develop slowly and are the last to mature .

Among the options listed , the cortex (especially the prefrontal cortex) is the most advanced part of the brain and is the last to mature . The frontal lobe , which is part of the cortex , is especially under developed at birth and develops gradually over the first few years of life .

Therefore , the least developed part of the brain at birth is the cortex (specifically the prefrontal cortex , which is part of the frontal lobe) , as it is responsible for higher cognitive functions like decision-making , planning , and self-control .

While the frontal lobe is part of the cortex , the cortex itself is the broad structural region responsible for complex functions and is the last to mature .

Between the choices , "cortex" (G) is the most accurate and comprehensive answer , as it reflects the overall development timeline â the cortex , especially its association areas , is the least developed at birth and develops over time .

Other structures like the brain stem and thalamus are functionally active at birth and develop early . The limbic system , hypothalamus , and basal ganglia develop earlier as well .

Thus , the least developed part of the brain at birth is the cortex .

<answer>G</answer>

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